

SE4020 – Mobile Application Development & Design – Assignment 2

Part A – iPadOS or macOS App

AI-Driven UI Design Process

Submitted by:

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I. AI tool selected and justification

Tool used: Claude (Claude Artifacts), model Claude Opus.

Why Claude was chosen	Detail
Renders runnable, interactive mockups not just static images	Claude Artifacts returns a self-contained HTML/CSS/JS screen I can open, resize to iPad dimensions, and screenshot. Unlike Midjourney/Galileo (which produce pretty but non-functional raster images), an Artifact lets me verify <i>layout behaviour</i> (a three-column split, a scrolling timeline, tappable chips) the things that matter for an iPad app.
Understands platform conventions in text	I could specify "iOS NavigationSplitView, SF Symbols, Dynamic Type, Human Interface Guidelines" in plain English and Claude reasons about them, rather than me hunting a visual style. This keeps the mockup honest to the platform I'm building for (SwiftUI).
Tight iteration loop	I can ask for a variation, critique it, and request a targeted change in seconds, all in one conversation ideal for generating <i>three deliberately different</i> directions and then synthesising.
Same vendor as my implementation assistant	The design vocabulary (colour tokens, spacing scale, component names) carries straight into the SwiftUI build, reducing translation loss between mockup and code.
Domain reasoning	Claude could weigh safeguarding context (RIDDOR incident order, allergy prominence, GDPR no-backdating) when laying out screens a childcare-specific judgement a generic image generator cannot make.

Why not the alternatives: Midjourney / Galileo AI produce non-editable images with invented, non-HIG controls; Figma AI is strong for high-fidelity polish but slower to spin up three *structurally* different directions; v0 by Vercel targets web/React conventions, not

iPadOS multi-column idioms. Claude gave the best balance of *platform-correct structure* + *iteration speed* for an iPad SwiftUI target.

II. Generated Mockups and Screenshots + Critical Evaluation

Variant A: Native split-view, calm cards

NurseryConnect

Sunny Days Nursery

Children

6 in your care

Insights

Trends & wellbeing

Incidents

1 open report

Children

Monday 1 June · 6 present

Q Search children

YOUR KEY CHILDREN

OB

Olivia Bennett

2y 4m · Sunbeams Room

<

>

NP

Noah Patel

3y 1m · Rainbows Room

>

AF

Amelia Foster

1y 6m · Acorns Room

<

>

LN

Leo Nakamura

2y 8m · Sunbeams Room

>

SR

Sophia Reyes

3y 4m · Rainbows Room

>

JT

Jack Thompson

1y 10m · Acorns Room

>

SM

Sarah Mitchell

KEYWORKER

NP

Noah Patel

3y 1m · Rainbows Room

Keyworker · Sarah Mitchell

In today · since 08:35

Photos consented

Activity

Meal

Nap

Nappy

Wellbeing

Report

Today's Diary

Monday 1 June 2026

Meal

Lunch — chicken & rice

11:30

Ate all, used his fork well.

All eaten

Logged by SM

Activity

Outdoor obstacle course

10:00

Confident climbing and balancing. Great gross-motor work.

Happy

Logged by SM

Report

Arrival & drop-off

08:35

Bouncy and chatty at drop-off.

Happy

Logged by SM

NurseryConnect

Sunny Days Nursery

Children

6 in your care

Insights

Trends & wellbeing

Incidents

1 open report

Insights

Development & wellbeing · this week (26 May – 1 Jun)

Olivia Noah Amelia Leo Sophia Jack

Average mood

Happy

Brighter than last week

Average nap

1h 19m

Steady · 5 of 5 days

Meals eaten

88%

Good appetite

Mood trend

Daily wellbeing check-ins · Amelia

Happy

Settled

Content

Unsettled

Mon

Tue

Wed

Thu

Fri

Nap duration

Minutes asleep per day

80

95

70

75

75

Mon

Tue

Wed

Thu

Fri

Meal portions eaten

% of lunch portion finished

90%

70%

100%

85%

95%

Mon

Tue

Wed

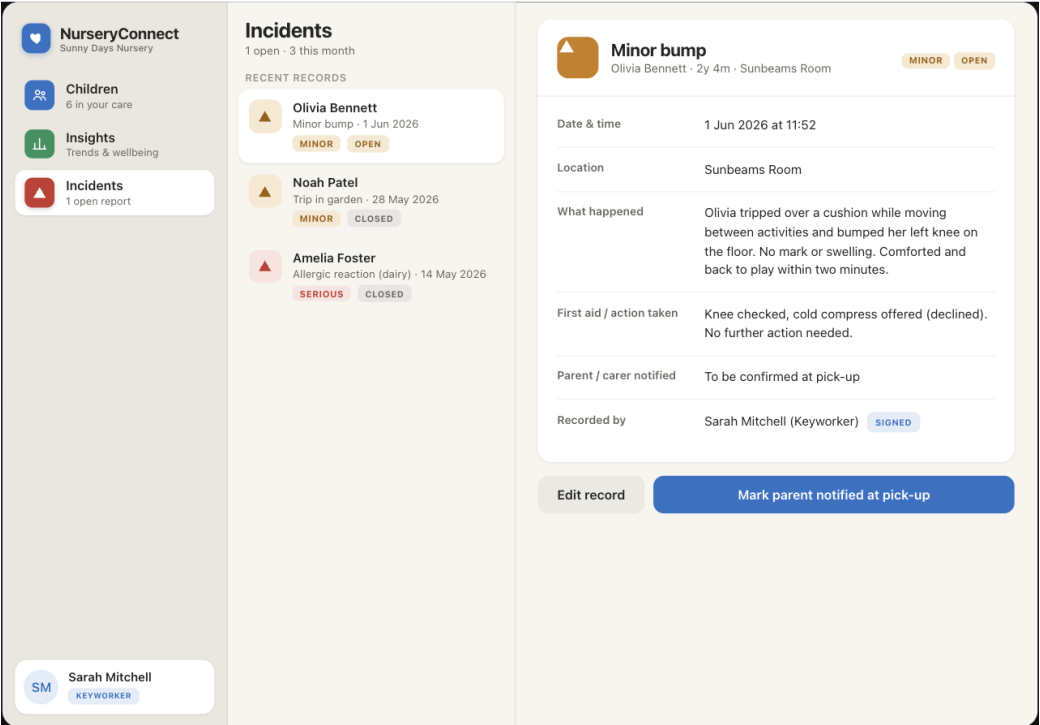
Thu

Fri

SM

Sarah Mitchell

KEYWORKER



Axis	Evaluation
Keyworker role	Strongest fit. The roster→child→day flow mirrors how a keyworker actually works (scan my children → open one → log/observe). Quick-log bar puts the highest-frequency action one tap away.
NurseryConnect context	Allergy/consent get dedicated, colour-coded prominence (red chips) without shouting; incident reporting is present but visually subordinate to daily logging correct weighting for a <i>daily</i> tool.
iPadOS conventions	A textbook NavigationSplitView sidebar + content + detail is <i>the</i> iPad idiom; persistent context across columns, no modal overload, scales to Stage Manager / different iPad sizes.
Weaknesses	The Insights dashboard is "one click away" rather than front-and-centre; risk of the detail column feeling busy if a child has many chips.

NurseryConnect
Sunny Days Nursery · Mon 1 Jun

All children

Olivia

Noah

Amelia

Leo

Sophia

Ja

Sarah Mitchell
Keyworker

SM

Home

Children

Alerts

Development & Wellbeing

All children - daily insights from this week's diary

Mon 26 May – Sun 1 Jun

Wellbeing mood trend

Brighter

Daily mood check-ins - scale 1 (upset) to 5 (very happy)

Date	Mood (1-5)
Mon 26	2
Tue 27	3
Wed 28	2
Thu 29	3
Fri 30	3.5
Sat 31	2.5
Sun 1	3

Nap duration

Minutes asleep per day

Date	Nap duration (m)
Mon 26	78
Tue 27	85
Wed 28	72
Thu 29	90
Fri 30	80
Sat 31	75
Sun 1	82

Meal portions eaten

% of lunch finished - none → some → most

Date	% of lunch finished
Mon 26	80%
Tue 27	62%
Wed 28	90%
Thu 29	95%
Fri 30	72%
Sat 31	88%
Sun 1	93%

Today's alerts

Keep these front of mind on the floor

Olivia Bennett
Peanuts allergy

ALLERGY

Amelia Foster
Dairy & Egg allergy

ALLERGY

Amelia Foster
No photography consent on file

CONSENT

Olivia Bennett
Vegetarian meals

DIET



NurseryConnect

Sunny Days Nursery · Mon 1 Jun

Sarah Mitchell

Keyworker

SM



Home



Children



Alerts

Children

Your 6 key children · all present today

OB

Olivia Bennett

2y 4m · Sunbeams Room

In 08:10

 Peanuts

 Vegetarian

 Photos

NP

Noah Patel

3y 1m · Rainbows Room

In 08:35

 Photos

AF

Amelia Foster

1y 6m · Acorns Room

In 07:55

 Dairy

 Egg

 No photos

LN

Leo Nakamura

2y 9m · Sunbeams Room

In 08:20

 Halal

 Photos

SR

Sophia Reyes

3y 4m · Rainbows Room

In 09:05

 Photos

JT

Jack Thompson

1y 10m · Acorns Room

In 08:45

 Photos

SM



NurseryConnect

Sunny Days Nursery · Mon 1 Jun

Sarah Mitchell

Keyworker

SM



Home



Children



Alerts

Incidents & Alerts

1 open report · safeguarding records this month

 Olivia Bennett

Minor bump · Sunbeams Room · 1 Jun 2026 at 11:52

MINOR

OPEN

>

 Noah Patel

Trip in garden · Garden · 28 May 2026 at 14:10

MINOR

CLOSED

>

 Amelia Foster

Allergic reaction (dairy) · Acorns Room · 14 May 2026 at 11:38

SERIOUS

CLOSED

>

SM

Axis	Evaluation
Keyworker role	Mixed. Great for a <i>reflective</i> end-of-week review, but a keyworker's primary job is logging during the day , not reading analytics leading with charts buries the most-used action (quick-log) behind a navigation step.
NurseryConnect context	Showcases the NEW Swift Charts feature beautifully and makes safety "alerts" glanceable; but daily diary logging the data <i>source</i> is demoted, which inverts cause and effect.
iPadOS conventions	A grid-of-cards dashboard is more of a web/macOS pattern; it under-uses the iPad's signature multi-column navigation and can feel like a ported web app.
Strengths to steal	The emoji-axis mood line, the none→amber→green meal colour logic, and the headline stat tiles are excellent and worth grafting.

Variant C: Single-column timeline, editorial minimal

Good morning, Sarah

Monday 1 June - your six key children are all in today. Tap a child to open their day.

- 
Olivia Bennett
 2y 4m · Sunbeams Room

 Peanuts
 >
- 
Noah Patel
 3y 1m · Rainbows Room
 >
- 
Amelia Foster
 1y 6m · Acorns Room

 Dairy, Egg
 >
- 
Leo Nakamura
 2y 9m · Sunbeams Room
 >
- 
Sophia Reyes
 3y 4m · Rainbows Room
 >
- 
Jack Thompson
 1y 10m · Acorns Room
 >

OB

Olivia Bennett

2y 4m · Sunbeams Room · Keyworker Sarah Mitchell

 Peanuts allergy  Vegetarian  Photos consented

Today Insights Incidents

TODAY · MONDAY 1 JUNE

- 15:10

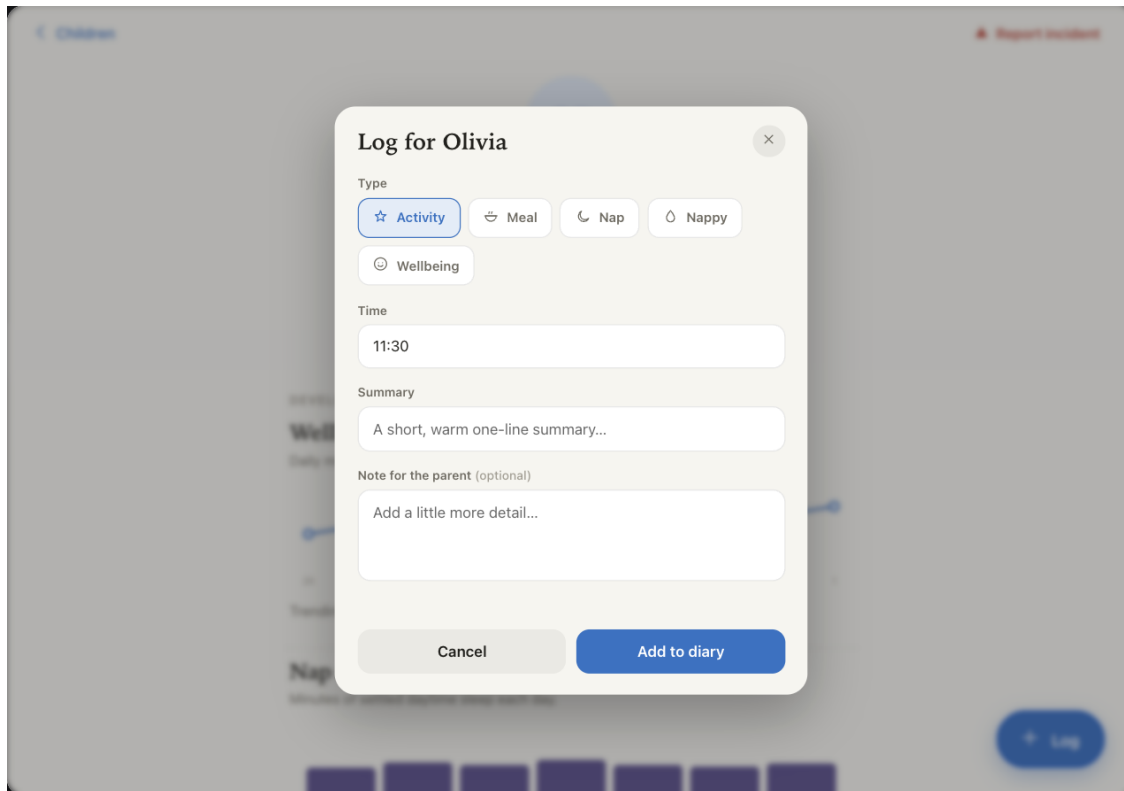
Built block towers with Leo
 Counted the bricks up to five — lots of giggles when it toppled.
 Happy · Activity · logged by SM
- 14:05

Nappy change
 Wet — changed and barrier cream applied. Skin healthy.
 Nappy · logged by SM
- 12:40

Napped for 1h 15m
 Settled quickly with her comforter.
 Settled · Nap · logged by SM
- 11:45

Lunch — pasta with tomato sauce

+ Log



Axis	Evaluation
Keyworker role	Lovely for deep focus on one child, but a keyworker constantly switches between children collapsing to one column forces a back-and-forth that the iPad's width was meant to eliminate.
NurseryConnect context	Minimal chrome de-emphasises safety signals (allergies become "subtle pills"), which is the wrong trade-off for a safeguarding tool danger info should be <i>more</i> prominent, not quieter.
iPadOS conventions	A single-column, phone-style layout with a floating action button wastes ~60% of the landscape canvas and reads as an iPhone app stretched onto an iPad.
Strengths to steal	The spacious, typography-led timeline rows and the calm "Things 3" whitespace improve readability worth adopting <i>inside</i> the detail column.

III. Final design decision and synthesis rationale

Selected **base:**
Variant A (Native split-view, calm cards) synthesized with the best elements of B and C.

Why **A** **is** **the** **spine:**
the three-column NavigationSplitView is the only variation that is simultaneously (a) true to keep the roster visible, drill into a child, never lose context; (b) correct for the safeguarding context allergies/consent get colour-coded prominence and incident reporting lives as a clearly-marked red affordance; and (c) genuinely iPadOS-native, using the full landscape canvas and scaling cleanly across iPad sizes and Stage Manager. B inverts the keyworker's real priorities (logging > reviewing), and C throws away the iPad's defining advantage (multi-column context).

What was synthesized in (not discarded):

Grafted element	From	Where it lives in the final build
Emoji-axis mood line chart, nap bar chart , none→amber→green meal-portion colour logic , and headline stat tiles (avg mood / avg nap)	B	The Insights dashboard (InsightsDashboardView) reached via the sidebar's "Insights" section rather than as the home, so daily logging stays primary.
Spacious, typography-led timeline rows and calm whitespace	C	The diary timeline inside the child detail column (DiaryEntryRow), giving readable rows without heavy chrome.
Three-column sidebar → list → detail , roster row design, quick-log bar, red incident affordance, allergy/dietary chips, keyworker footer	A	The whole app shell (RootSplitView, ChildDetailView, QuickActionBar).

Final design decisions traceable to the build:

- **Structure:** RootSplitView sidebar (Children / Insights / Incidents) → content list → detail, exactly as in Variation A.
- **Palette:** soft blue #3D7DCA, warm off-white #F7F6F2, white cards, with **red reserved exclusively for incidents/allergies** encoded as semantic tokens in AppColors and backed by light/dark colour sets (Dark Mode for free).
- **Safety prominence:** allergy and photography-consent flags surface in red on both the roster row and the child header card the synthesis deliberately keeps these *loud*, rejecting Variation C's "subtle pills".
- **Daily-first ordering:** quick-log actions sit at the top of the day view; the Insights dashboard (the new Swift Charts feature) is one deliberate navigation step away honouring the keyworker's real frequency of use while still showcasing the advanced library.
- **Platform feature carried through:** the split-view backbone and a PencilKit signature on the full-screen incident form (Variation A's "Report Incident" affordance taken to completion).

Outcome: the shipped UI is *Variation A's architecture, hardened with B's charts and C's typographic calm* an iPad-native, safeguarding-aware, daily-logging-first tool for the Keyworker, rather than any single AI mockup adopted wholesale.